U.S. Sporting Goods Retail Landscape

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Introduction

Since 2020, the sporting goods sector has transitioned from a steady retail staple to one of the most resilient growth engines in the global consumer market. While the 2010s were defined by the "athleisure" boom, the initial integration of performance wear into daily life, the post-pandemic era has catalyzed a deeper cultural shift. Sports have evolved beyond mere physical activity to become the de facto social infrastructure of this decade.

From the unprecedented global viewership of the 2022 World Cup and 2024 Paris Olympics to the record-breaking participation rates in local marathons and social-club sports like pickleball, physical engagement is now a primary driver of social status and community.

The retail landscape has moved in lockstep with this behavioral evolution. Supporting a franchise or adopting a new sport now requires specialized items, turning equipment and apparel into essential spending for many. According to Mintel's *Sporting Goods Retailing – US – 2024* report, this engagement is remarkably strong as 39% of consumers express a willingness to purchase new seasonal gear even when their current equipment remains functional. This shift highlights a transition from a "need-based" replacement cycle to a "trend-based" refresh cycle, providing retailers with consistent, year-round revenue opportunities.

This analysis evaluates the current state of the $110 billion+ U.S. sporting goods market, identifying the market leaders and the specific operational levers driving their dominance. It is structured to provide up-to-date insights for managers and investors on the operational decisions that will define future growth opportunities.

Data Understanding

I worked with the *Spend Patterns* dataset from SafeGraph, which provides anonymized monthly credit and debit transaction data at specific points of sale, referred to as Points of Interest (POI) within the dataset. Because the data encompasses a broad spectrum of retail transactions, it is naturally extensive and requires targeted filtering to isolate records relevant to the sporting goods sector.

To ensure the analysis accurately reflects current performance and supports future strategic recommendations, I focused exclusively on the 2025 dataset (covering January through November). Prioritizing the most recent data ensures the highest market relevance while optimizing processing efficiency.

The following columns were utilized to conduct the analysis:

SUB_CATEGORY: The label associated with all 6 digits of the NAICS code, used to find targeted category.

LOCATION_NAME: The name of the POI. In practice, this column was more useful in identifying a specific brand than the column BRANDS, thus used for retailer grouping.

PLACEKEY: A unique and persistent ID tied to each POI, used to pinpoint specific store location.

RAW_TOTAL_SPEND: The total amount spent at POI in transactions captured, used for sales aggregation.

REGION: The state, province, county, or equivalent, used for geographic grouping.

SPEND_DATE_RANGE_START: The start time for the measurement period, used to aggregate transactions into months.

In this analysis, RAW_TOTAL_SPEND is used as the primary metric for determining market trends and business profitability.

Data Preparation

Filtering

The original dataset consisted of transactions from all retail categories in the US, so the first filter was applied during the downloading process by selecting only “Sporting Goods, Hobby, and Musical Instrument Stores” for the TOP_CATEGORY field. As the name suggests, the data provider grouped Sporting Goods with two other leisure and art subcategories, which I believe was an old way of classifying that does not accurately reflect how popular sports products have become. Regardless, this filter gave me a subset as the starting point.

I performed the next filter on the SUB_CATEGORY field to pick out “Sporting Goods Stores” from the rest. The result was a list of all specialized stores that sell sports equipment, including small independent businesses. At this step, it was important to set the scope and segment for the analysis.

From personal experience and observation, I decided that the top five retailers by store count should be sufficient to reflect the overall picture of the sporting goods market. I sorted the retailers by total locations, and the first five results were Hibbett Sports, Dick's Sporting Goods, Big 5 Sporting Goods, Champs Sports, Academy Sports + Outdoors, which together comprised 26,893 records of the year.

Data Cleaning

I looked for duplicates in the dataset and found 2,391 repeated records, which I removed using drop duplicates method.

Examining the dataset, there were some columns with considerable missing values. The two with the highest numbers were PARENT_PLACEKEY, which is the unique ID of a parent location, and CUSTOMER_HOME_CITY, which provides the customer’s estimated home location. After assessing potential use cases, I determined that these two features would not play a role in the analysis and could be completely dropped.

The next two columns with large missing values were SPEND_PCT_CHANGE_VS_PREV_MONTH and SPEND_PCT_CHANGE_VS_PREV_YEAR, presenting the percent difference in total spending at a POI between that record and its last month's / last year same-month's. By definition, these percentages could only be calculated if their respective historical data were available. Therefore, it was normal not to have all values available since replacing the Nulls with anything would distort the fields' intended purpose.

For the last two columns with the least missing values, I opted for replacing the Nulls with their respective PLACEKEY's matching values. The BUCKETED_CUSTOMER_INCOMES gives the distribution of estimated customer income buckets, and MEAN_SPEND_PER_CUSTOMER_BY_INCOME provides the average spending by customer income buckets. I realized that these values were dependent on each POI's customer base, thus their monthly aggregated statistics could not be much different given a short period of time, in this case one calendar year.

The following step was actually performed after further analysis down the road. Locations that did not have complete eleven months of data tended to have integrity issues. For example, some stores only showed up a single time in the middle of the year, or they recorded unreasonably low revenues throughout. As a result, without additional information and to prevent skewing the analysis, I removed all locations that had missing months.

Finally, using boxplots with the Interquartile Range method, I looked at outliers in the RAW_TOTAL_SPEND column and found two extreme values from one location of Academy Sports + Outdoors. The numbers were recorded in October and November, jumping a few hundred times over previous months' spending. I considered them as errors and removed from the dataset. After the removal, there were still outliers beyond the upper fences, but I perceived those as valid spendings from flagship stores or distribution centers. As a result, I ended up with a master dataset for the analysis.

Exploratory Data Analysis

Overall Trend

For the first part of EDA, I wanted to understand the overall trend and seasonality pattern of the sporting goods retail market. I grouped total spending by retailer and month, making a line chart to display the aggregated data.

Three insights could be immediately drawn from this chart:

- The current market is not a level playing field, at least by sales numbers. Dick's Sporting Goods and Academy Sports + Outdoors had their sales several times higher than competitors' at any point during the year. In some peak demand months, the two accounted for over 80% of total sales from our top five retailers. It came as a surprise for Academy, considering they had the fewest stores.
- Spring and Summer months are the high season for sporting goods. Probably not new to industry experts, but to outsiders March-August period is the opposite of general apparel that sells more towards late Summer and end-of-year holidays.

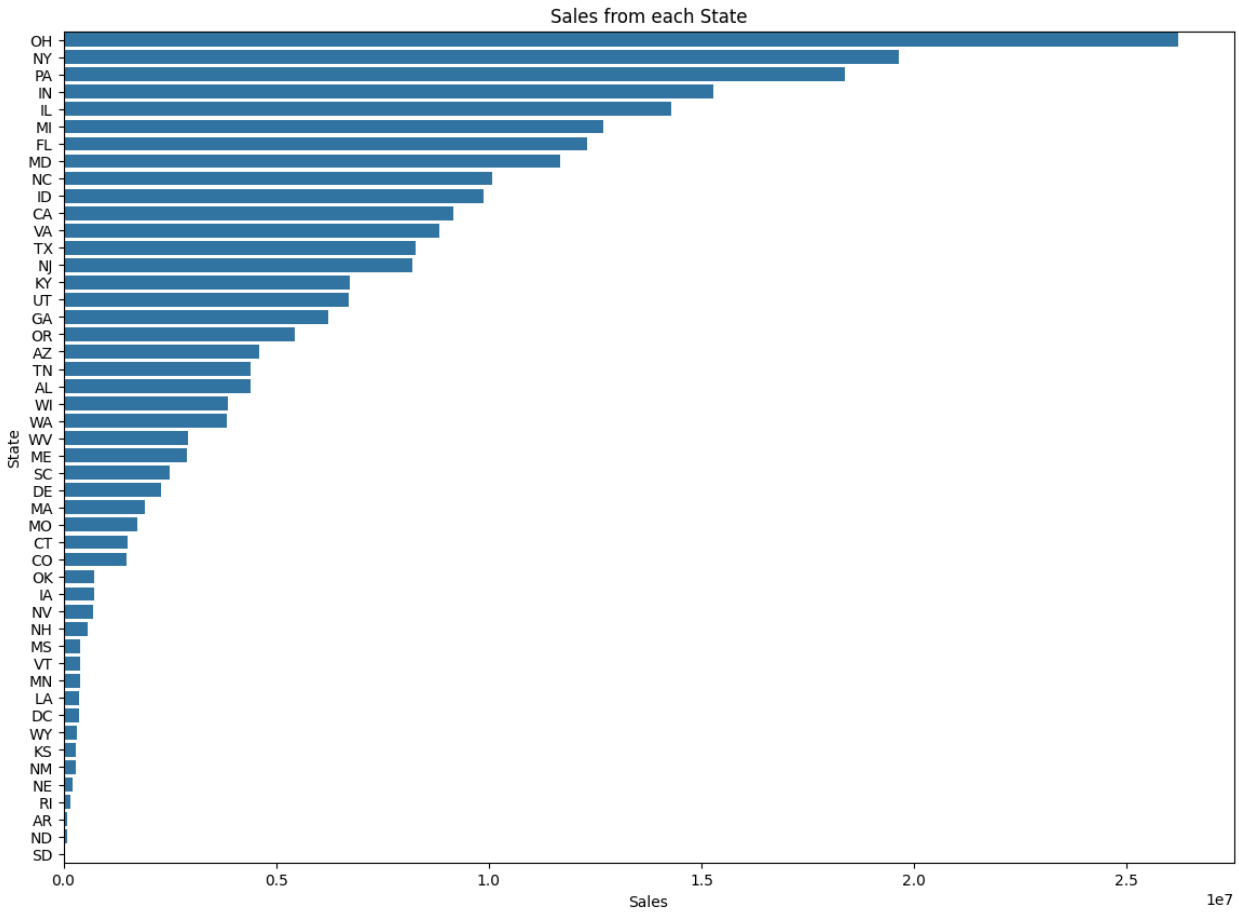
- Seasonality has a substantial impact on retailers that offer a wide range of products. The contribution was clearly important to Dick's annual sales, while it played a smaller role on Academy. The exception was Big 5.

From the above initial overview, I identified Dick's Sporting Goods as the principal retailer of this market. Not only did they carry an extensive store count of 789 locations, second to Hibbett Sports, the company also successfully translated that physical presence to market domination. In every month of 2025 until November, their revenue was more than double that of the closest rival, Academy. Each Dick's location also generated much higher sales than any other competitor, showing a remarkable level of efficiency in operations.

With that clear distinction between Dick's Sporting Goods and the rest, it was logical to view them as the case study for success. For the next part of EDA, I would dive deeper into analyzing the company.

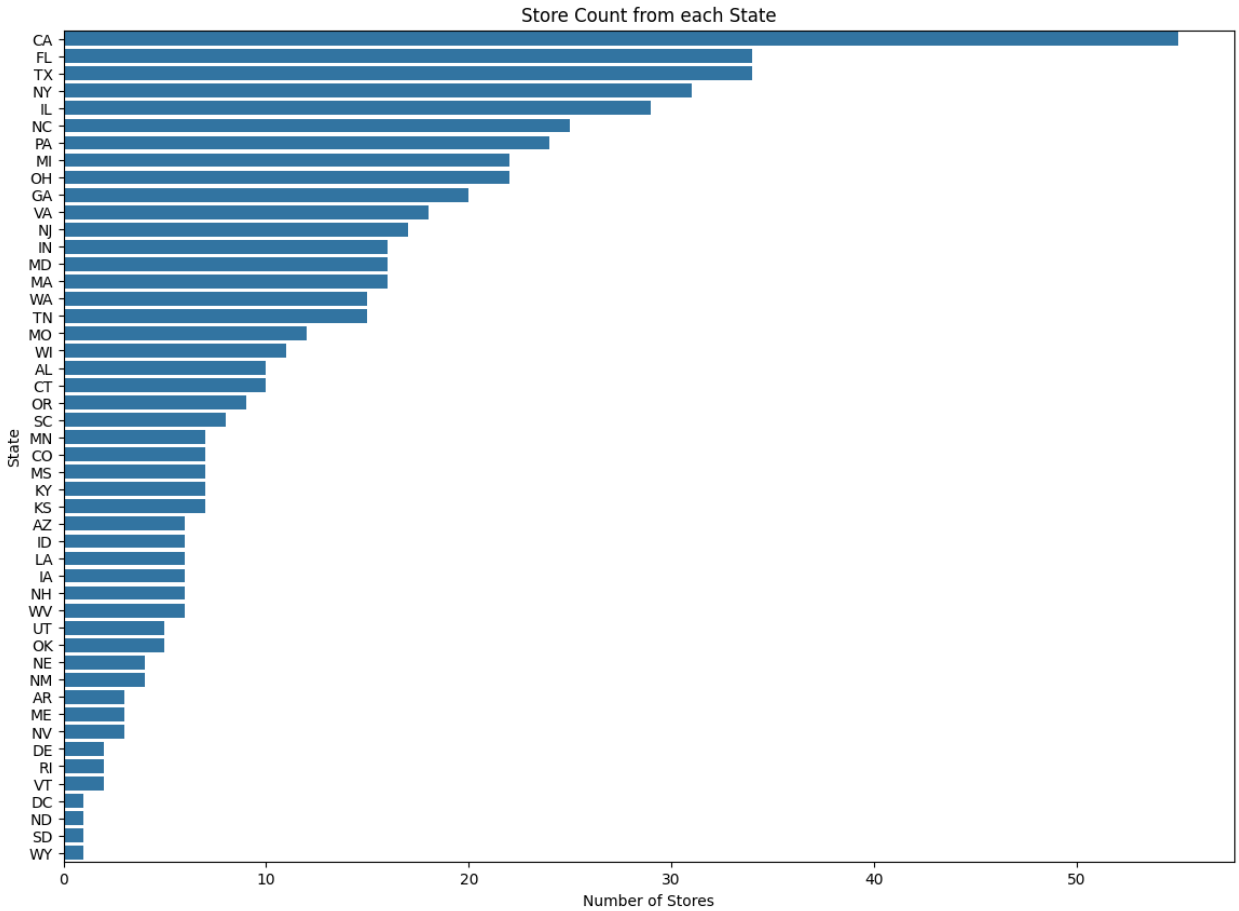
Case Study – Dick's Sporting Goods

The first thing I wanted to look at was sales distribution by state. If there were concentrated revenue from any regional market, we would be able to identify it and hopefully find out the background strategy. I used a barplot chart to show side-by-side total revenue from each state in 2025.



Three states emerged from the chart as the main drivers of Dick's income: Ohio, New York, and Pennsylvania, in order of highest sales. It might not be a surprise to see the latter two since the company was founded in New York and its headquarters were moved to Pennsylvania in 1994, but Ohio appearing as number one in sales was an interesting discovery. I went one step further and reviewed the company's annual report to see if there was any shipping center in this area that could be incorrectly reflected in the transaction data. In fact, there was no significant distribution facility in Ohio that might influence the number.

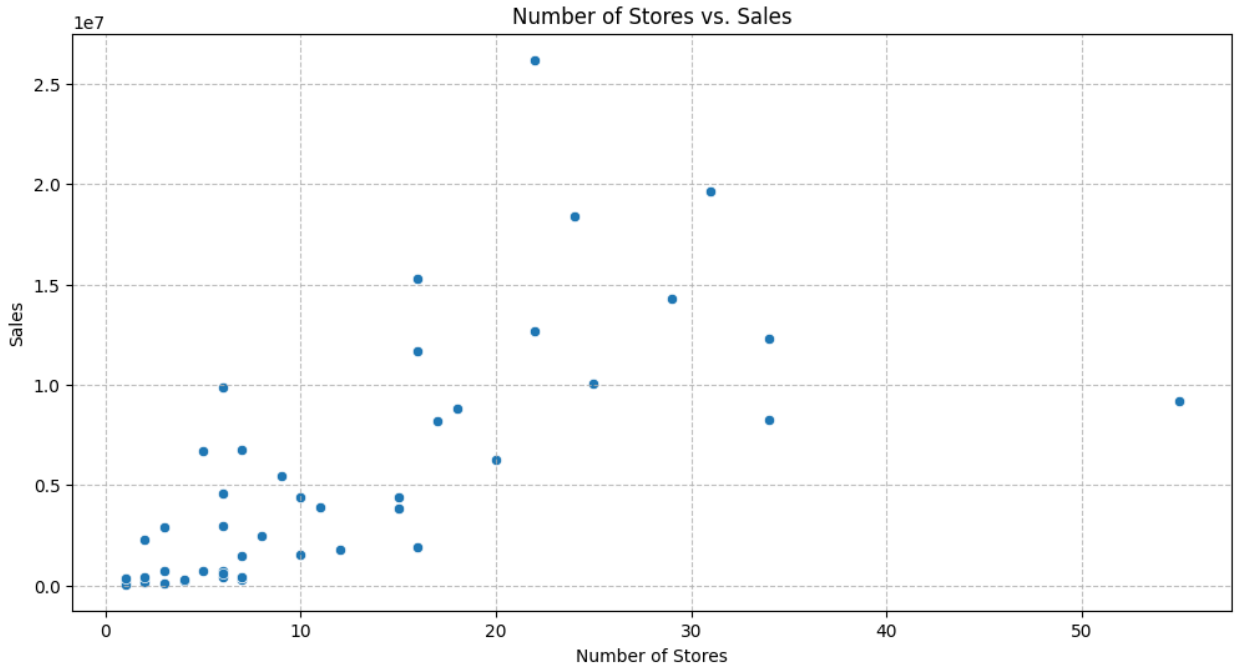
With that piece of information, I next took a look at the number of stores in each state. The reasoning was that if store counts gave the same pattern, it could hint that more stores generally resulted in higher total sales. Another barplot chart was created for the store count from each state.



A different hierarchy appeared, which was actually aligned with the common thinking that is states with higher population have more retail locations. California, Florida, and Texas made the top three, while Ohio was in ninth position.

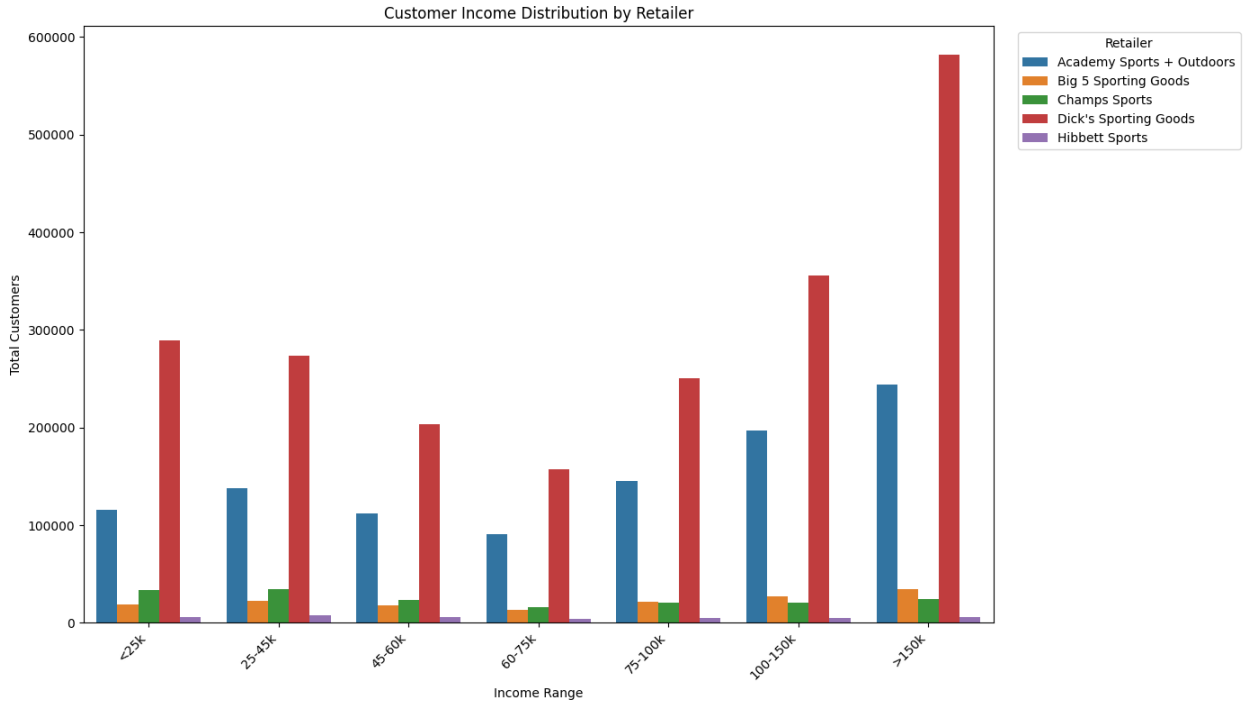
With information from sales and store count distribution, the data seemed to suggest an insight – number of stores is not a good indicator for profitability. There were two hypotheses to follow up on: either that customers in some states have a tendency to spend more at Dick's locations, or the company has not successfully captured a few major state markets.

In the scope of this analysis, there was not enough marketing and demographic data to explore these possibilities. Therefore, I continued with examining the relationship between store number and revenue. A good way to see the interaction between two features is a scatterplot, so I made one with number of stores on x-axis and sales on y-axis.

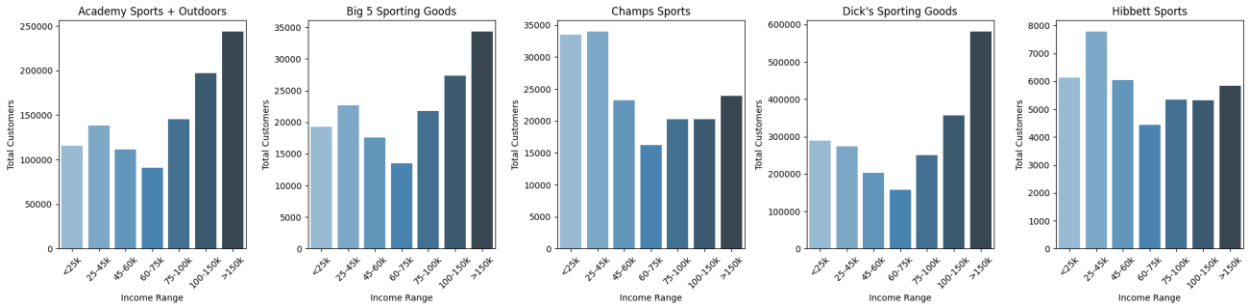


As predicted, the graph did not hint at any straight line for a strong relationship and instead looked more like a cone-shaped pattern. I came to the first conclusion**: rather than just being omnipresent**, **there must be different factors that contributed to a retailer's high revenue.**

Since the dataset included just a few demographic-level data, the only meaningful feature to assess was customer income level, which was in the BUCKETED_CUSTOMER_INCOMES column. Data were aggregated into a string of dictionary-type pairs, of which the key was the income range and the value was the number of customers belonging to that range. I parsed through each row, put the values into respective new columns that reflect the incomes, and then transformed the grouped table to a long format for visualization. My goal was to look for differences in customer income distributions between Dick's and the others, since it could potentially reveal distinctions in their strategies.



After summing up the income columns, I visualized all retailers' distributions side by side to review the market dynamics. Once again, the chart confirmed the dominance that we saw earlier in the analysis, with Dick's Sporting Goods leading the measurement by almost doubling their closet competitor at any income range. I made another five subplots, one for each retailer, on the same scale to better assess their patterns.



We got two groups of distributions: Champs Sports and Hibbett Sports made up the first group with more count at low ranges, while the rest made up the second group with more count towards high ranges. This separation pointed to the fundamental differences in customer segments targeted by these retailers, either leaning towards youth and low-income customers, or focusing on those with more money to spend. It's also worth noting that the "60-75k" range was the

lowest at every retailer, which may suggest that shoppers with this income level are very conscious of their budget and sporting goods is not a convincing product category for them.

Looking at the second group, we could see a slight differentiation from Dick's. Although all players in this group attracted high-income customers, Dick's demonstrated a significant share of ">150k" compared to other ranges. The insight implied that the company had successfully positioned itself as a trusted premier sporting goods retailer, especially to those buyers who had spare income to spend on their purchases. I arrived at the second conclusion: **with an overall strategy of offering premium products and services, Dick's Sporting Goods has become the one national brand that not only effectively captivates high-income customers but also appeals to broader buyers who care about the shopping experience**.

Indication and Recommendation

The analysis gives a clear picture of the sporting goods retail market in 2025. Largest retailers all have over 300 locations across the country, with the top two each owning around 800 stores. For revenue, Dick's Sporting Goods is the winner by a wide margin, targeting high-income customers through an omnichannel retail and premium services strategy.

Higher store count does not always translate to more sales, though probably important for efficient logistics and brand visibility. Sales are more concentrated in some states than other but not necessarily related to population or income. The high season for sporting goods is Spring and Summer, aligning with suitable temperatures and the start of the school year.

My first recommendation for managers would be to put more focus on middle-income and early career customers. This group seems to be underserved by the market and could have found their needs from other buying options like e-commerce, which is often more competitive in prices. If left unattended, these customers would get familiar with the alternatives and not develop a relationship with retailers at all.

The next recommendation is that, in order to be able to compete with the giants, smaller brands need specific differentiation strategies to set themselves apart from competitors. In a market that is saturated with similar products and competitive pricing, it is crucial to offer something unique beyond just transactional activity. Whether it is good customer service or seamless reverse logistics, the key is to find something that your customer segment values the most.

Last but not least, for investors, I suggest to really consider the trade-off between running a large number of stores nationwide and concentrating on a few strategic regions. As seen in the analysis, even the biggest player does not make the most revenue out of its highest-store-count states. Territory may play a notable role in sporting goods retail, just like in many other consumer goods. With

a precise strategy, any company could optimize resources to maximize its competitive advantages.

What about a proposal strategy for a new entrant to this market? It would be a consolidation of the three recommendations above – target the middle-income segment that is being underserved, start out in a few regions that are historically up for competition, and develop a compelling value proposition aligning with customer interest.

Acknowledgement of Limitations

- The monthly revenue of Dick's ~$30m did not match the company's 10-K. It was probably due to data was collected from credit and debit card transactions only, which is a subset of all payment types.
- The top 5 by store count is not the top 5 by sales. The reason to pick top 5 for store count is brand visibility and awareness.
- There are potentially other retailers owning subsidiaries that when combined would be in the top 5. However, to the typical consumer, they are different brands and thus treated as separate entities in this analysis.
- December data was not available, which could change the assumption on seasonality.